



Quiz No. 2

CS3004

Software Design and Analysis

Total marks: 30

Total time: 40 minutes

Date: 21/04/2026

Reg. No. _____

Name. _____

Scenario: Bookstore Checkout System

An online bookstore allows customers to purchase books using a ShoppingCartPage. The system stores OrderDetails (including items, quantities, and prices) and CustomerAccount information such as the user's saved address and payment method. When a customer clicks the "Checkout" button, the ShoppingCartPage directly subtracts the order total from the CustomerAccount balance. Separately, when an OrderDetail record shows that a book is out of stock, that same OrderDetail record automatically cancels the entire customer order without notifying the customer. Over time, the store manager notices serious problems: customers' account balances are deducted even when an order fails due to stock issues, orders are cancelled without any record of who initiated the cancellation, and debugging the checkout process is nearly impossible because multiple components modify order and account data independently without any central coordination.

- Identify three analysis classes from the scenario. Classify each as Boundary, Control, or Entity.
- Justify each classification in one sentence using the definition of each stereotype.
- Draw a robustness diagram showing the actual (flawed) interactions in the scenario. Include the actor and all analysis classes.
- Mark each communication violation on your diagram. Below it, list each violation and name the rule being broken.
- Introduce a new control class to fix the design.
- Redesign the interactions and redraw the robustness diagram following correct ECB rules.

SOLUTION

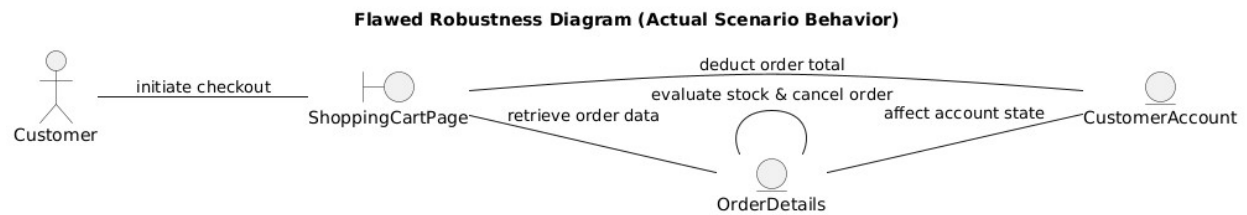
a) Identify three analysis classes with classification

- ShoppingCartPage → Boundary class
- CustomerAccount → Entity class
- OrderDetails → Entity class

b) Justification (one sentence each)

- ShoppingCartPage (Boundary): It represents the interface through which the user interacts with the system (UI layer).
- CustomerAccount (Entity): It stores persistent business data such as balance, address, and payment information.
- OrderDetails (Entity): It holds core transactional data about orders, including items, quantities, and prices.

c)



d) Communication Violations

1. Boundary → Entity direct interaction
 - ShoppingCartPage directly modifies CustomerAccount
2. Entity controlling logic
 - OrderDetails cancels the entire order
3. Lack of Control class
 - No central coordination of checkout process
4. Untracked actions
 - Order cancellation has no actor/control involvement

e) Introduce a new Control Class

New Control Class: CheckoutController

Responsibility:

- Handle checkout process
- Validate stock
- Deduct balance only after success
- Manage order cancellation properly
- Log actions

f)

